Reliability Assessment of hBN-Encapsulated TMD Devices from FETs to Inverters

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***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

As device miniaturization continues, the performance of traditional logic devices based on bulk semiconductors becomes increasingly constrained, either by the exacerbation of short-channel effects when the channel thickness is not scaled accordingly, or by the associated mobility degradation when it is [3]. To overcome this inherent trade-off of bulk channels, two-dimensional (2D) materials, whose carrier mobility exhibits weak thickness dependence, have attracted significant attention. Among them, transition-metal dichalcogenides (TMDs) offer suitable bandgaps and currently represent the backbone of active logic devices based on 2D materials [1].

However, important limitations arise with the use of 2D materials. While their dangling-bond-free interfaces mitigate mobility degradation, the absence of covalent bonding at the oxide–channel interface results in increased instabilities compared to the well-established SiO_2/Si interface [2].

II. Methods and Fabrication

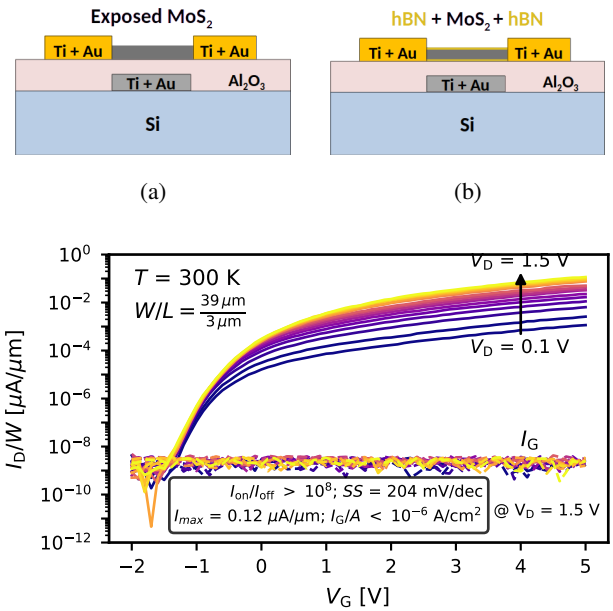

Fig. 1. **Cross-section schematics**: (a) exposed and (b) hBN-encapsulated MoS_2 FETs. (c) **Performance of the fabricated devices**: $I_{D(G)}$-V_G curves for different drain voltages and the main device metrics.

Acknowledgment

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks ...". Instead, try "R. B. G. thanks...". Put sponsor acknowledgments in the unnumbered footnote on the first page.

References

[1] Saptarshi Das et al. "Transistors based on two-dimensional materials for future integrated circuits". en. In: *Nature Electronics* 4.11 (Nov. 2021), pp. 786–799. ISSN: 2520-1131. DOI: 10.1038/s41928-021-00670-1. URL: https://www.nature.com/articles/s41928-021-00670-1 (visited on 03/31/2026).

[2] Hao-Yu Lan et al. "Reliability of high-performance monolayer MoS2 transistors on scaled high-k HfO2". en. In: *npj 2D Materials and Applications* 9.1 (Jan. 2025), p. 5. ISSN: 2397-7132. DOI: 10.1038/s41699-025-00527-7. URL: https://www.nature.com/articles/s41699-025-00527-7 (visited on 11/24/2025).

[3] Yuan Liu et al. "Promises and prospects of two-dimensional transistors". en. In: *Nature* 591.7848 (Mar. 2021), pp. 43–53. ISSN: 0028-0836, 1476-4687. DOI: 10.1038/s41586-021-03339-z. URL: https://www.nature.com/articles/s41586-021-03339-z (visited on 06/09/2025).

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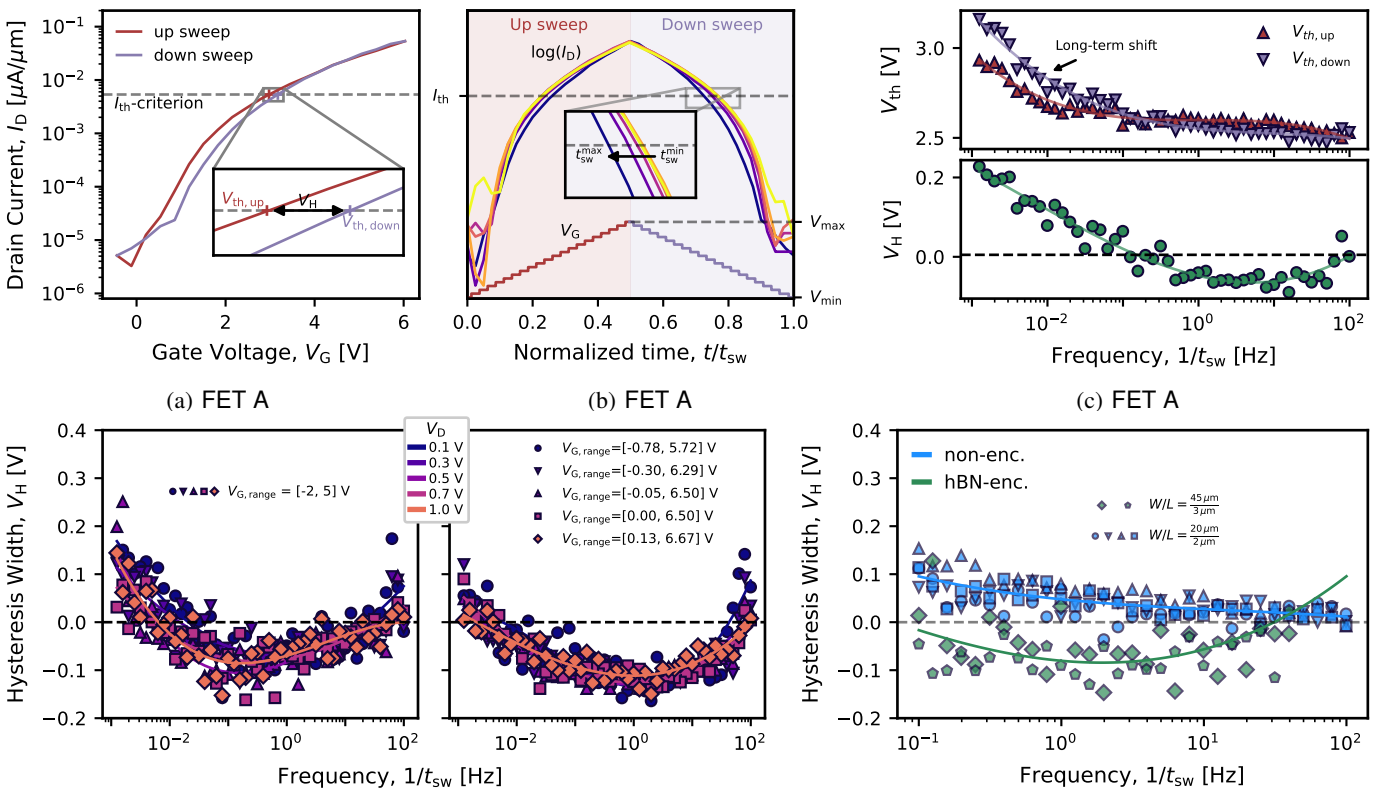
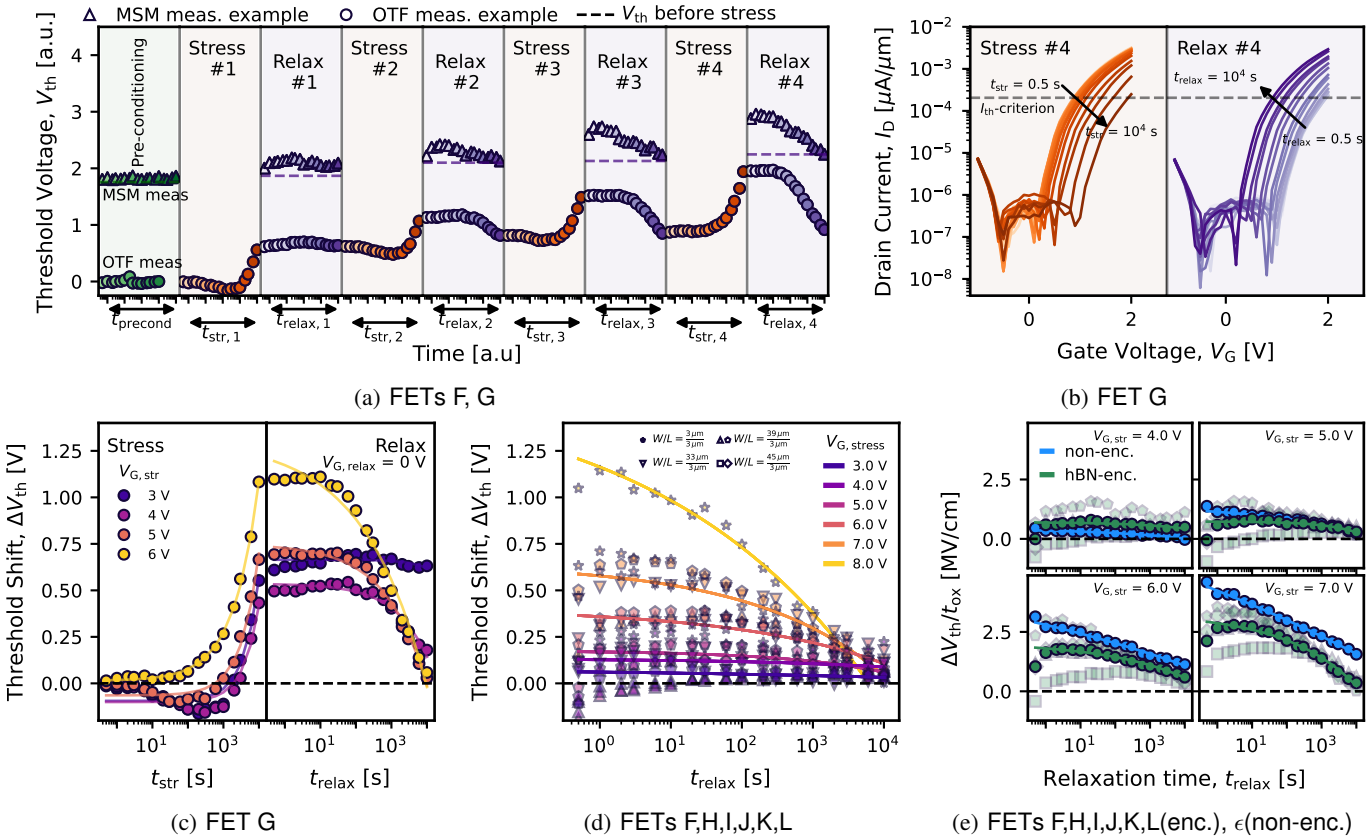


(a) FET A (b) FET A (c) FET A

(d) FETs B, C (e) FETs D, E(enc.), $\alpha, \beta, \gamma, \delta$(non-enc.)

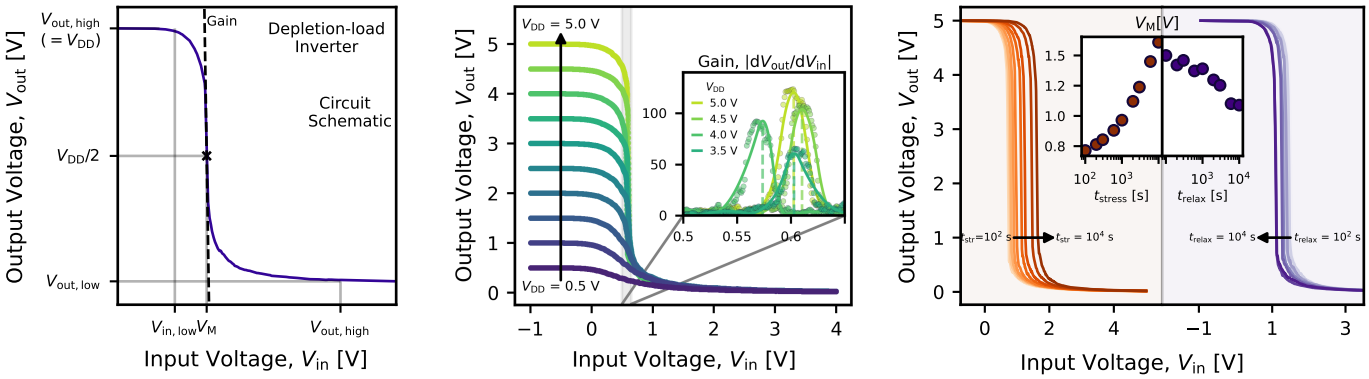
Fig. 2. (a) **Typical hysteresis curve**: the hysteresis width, V_H, is defined as the difference between the threshold voltages extracted during the up and down sweeps, i.e., $V_H = V_{th,up} - V_{th,down}$ evaluated at a constant current criterion I_{th}. (b) **Hysteresis dependence on the gate voltage sweep rate**: $\log(I_D)$ and V_G are shown as a function of the normalized sweep time, t/t_{sw}, for different sweep times, t_{sw}. (c) **Hysteresis-$1/t_{sw}$ curves**: Threshold voltages, $V_{th,up}$ and $V_{th,down}$, and hysteresis width, V_H, are shown as a function of the sweep frequency, $1/t_{sw}$. (d) **Hysteresis for varying V_D**: The hysteresis is independent from the applied drain bias, as evaluated on two different devices. More consistent hysteresis results are obtained when using an adaptive scheme for the gate voltage range targeting the same overdrive and the same sweep range in the subthreshold (right), as compared to a fixed gate voltage range (left). (e) **Hysteresis on encapsulated vs. non-encapsulated devices**: The hysteresis width on encapsulated devices is negative (counter-clockwise) and on non-encapsulated devices it was of a similar magnitude but positive (clockwise) evaluated at $V_D = 0.1V$.



(a) FETs F, G (b) FET G

(c) FET G (d) FETs F,H,I,J,K,L (e) FETs F,H,I,J,K,L(enc.), ϵ(non-enc.)

Fig. 3. (a) Example of a hysteresis curve and the corresponding time evolution of the gate voltage. (b) Comparison of the $I_{ds} - V_{gs}$ curves for the RO stage and the asymmetric stage.



(a) Inverter A (b) Inverter A (c) Inverter B

Fig. 4. (a) Example of a TDDB curve and the corresponding time evolution of the gate voltage. (b) Comparison of the $I_{ds} - V_{gs}$ curves for the RO stage and the asymmetric stage.